25 September 1974

MILITARY SPECIFICATION

TRANSMITTER GROUP, SIMULATED RADIAC, HOT SPOT OT-52( )/T

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. - This specification covers the Transmitter Group, Simulated Radiac, Hot Spot OT-52/T. The OT-52 is an RF transmitter and antenna which simulates a radioactive fallout pattern with radio waves. The OT-52/T consists of the three Transmitter, Simulated Radiac, Hot Spot T-1255/T, one Case, Simulated Transmitter CY-7349/T, one Antenna Assembly AS-2919/T and miscellaneous cables and support components.

2. APPLICABLE DOCUMENTS

2.1 Documents. - The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-585	Box, Wood, Wirebound
PPP-B-601	Box, Wood, Cleated-Plywood
PPP-B-621	Box, Wood, Nailed and Lock-Corner
PPP-B-636	Box, Fiberboard
PPP-F-320	Fiberboard, Corrugated and one Solid, Sheet Stock Container Grade and Cut Shapes
PPP-P-291	Paperboard, Wrapping, Cushioning
PPP-T-45	Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, Water Resistant, for Carton Sealing
QQ-S-781	Steel Strapping, Flat

FSC 6665

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MIL-P-116	Preservation, Methods of
MIL-C-3098	Crystal Unit, Quartz
MIL-P-11268	Parts, Materials and Processes Used in Electronic Communication Equipment
MIL-M-13231	Marking of Electronic Items

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-810	Environmental Test Methods

DRAWINGS

ELECTRONICS COMMAND

DLSM-B-748195	Transmitter Group, Simulated Radiac, Hot Spot OT-52/T
SM-A-678029	Burn-In Procedure for Transmitter, Simulated Radiac, Hot Spot T-1255( )/T

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(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 First article. - When specified in the contract or purchase order, the contractor shall furnish first article units in accordance with 4.3.

3.2 Construction. - The Transmitter Group, Simulated Radiac, Hot Spot OT-52/T shall be constructed in accordance with the requirements of this specification, and of Drawing DLSM-B-748195.

3.3 Parts, materials and processes. - In addition to any requirements of this specification covering parts, materials and processes, such items shall conform to MIL-P-11268, including the selection requirements therein.

3.4 Marking. -

3.4.1 General. - Marking shall conform to MIL-M-13231.

3.4.2 Serial numbers. - Each equipment shall bear a serial number in accordance with the requirements of MIL-M-13231.

3.5 Interchangeability. - Like units, assemblies, subassemblies and replaceable parts shall conform to requirement 7 of MIL-STD-454. (See 4.6)

3.6 Performance Characteristics. -

3.6.1 Transmitter, Simulated Radiac, Hot Spot T-1255/T. -

3.6.1.1 Operating frequency and stability. - The operating frequency shall be within 138 to 144 MHz and shall be crystal controlled. At variations of input voltage and power output specified in 4.11.1, the transmitted frequency shall not vary more than ± 2.5 KHz. The above criteria shall be met over the entire band utilizing available Military Standard (MS) crystals selected from MIL-C-3098/60. (See 4.11.1.1)

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3.6.1.2 Power output and stability. - The transmitter shall have a continuously variable power output. At room ambient temperature and input voltages specified in paragraph 4.11.1, the maximum power output shall be at least 225 mW but not more than 275 mW. When set on minimum power, the power output shall be less than 20 mW. (See 4.11.2)

3.6.1.3 Operation in different modes. -

3.6.1.3.1 Mode 1. - When operating in Mode 1 the transmitter shall be capable of automatically turning on after a selectable period of time between 1/2 hour and 6 hours and then automatically turning off after a selectable period of operation of between 1/2 and 6 hours. Both turn on and turn off times shall be selectable in 1/2 hour increments. (See 4.11.3.1)

3.6.1.3.2 Mode 2. - When operating in Mode 2 the transmitter shall be capable of automatically turning on at minimum power, after a selectable period between 1/2 to 6 hours in 1/2 hour increments. Following power turn on, the power shall increase to full power in 30 minutes. The power shall then decrease to minimum power in 4 hours. At that time the transmitter shall turn off automatically. (See 4.11.3.2)

3.6.1.3.3 Continuous wave (CW). - When operating in the CW mode the transmitter shall produce a CW RF signal capable of being continuously varied by the power control. When the power control is in position "0" the output shall be less than 20 mW; in position "10" the output shall not be greater than 250 mW ± 10%. (See 4.11.3.3)

3.6.1.4 Weight. - The weight of the transmitter without batteries shall not exceed 8 pounds. (See 4.8.1)

3.6.1.5 Battery Life. - The transmitter shall be capable of full power output for 12 hours at 0 ± 2.5°F in the CW mode without changing batteries. The power output shall be within 250 mW ± 10% after 12 hours. (See 4.9)

3.6.2 Antenna assembly AS-2919/T. -

3.6.2.1 Beam width. - The radiation pattern of the directional antenna shall have a beam width of not less than 20 degrees and not more than 50 degrees. (See 4.10)

3.6.3 Case, Simulated Transmitter CY-7349/T. -

3.6.3.1 Weight. - When loaded with all antenna and miscellaneous components and three transmitters T-1255/T (i.e., a complete Transmitter Group OT-52/T as per DLSM-B-748195) the carrying case assembly shall not exceed 135 pounds. (See 4.8.2)

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3.7 Environmental. - The Transmitter T-1255/T shall meet the performance requirements specified below under any combination of these service conditions.

3.7.1 Temperature. -

3.7.1.1 Low Temperature. - The transmitter shall be subjected to the low temperature extremes of 0 ± 2.5°F operating and -60 ± 2.5°F storage and shall meet the requirements of 3.6.1.2 when tested in accordance with 4.13.5.1.

3.7.1.2 High temperature. - The transmitter shall be subjected to the high temperature extremes of 125 ± 2.5°F operating and 160 ± 2.5°F storage and shall meet the requirements of 3.6.1.2 when tested in accordance with 4.13.5.2.

3.7.2 Electromagnetic interference (EMI) characteristics. - The transmitter shall meet the requirements of MIL-STD-461, for Methods CE06 and RE02 except as modified in Table 1. (See 4.14)

Table 1 Deviation from MIL-STD-461. -

CE06	RE02
Key Up X Key Down X	Narrowband fo Max Dev. 38 dBuV/M Other Freq Max Dev. 12 dBuV/M Broadband X

NOTE a. "X" denotes compliance with limits of MIL-STD-461.

b. Any deviation listed is added to the specified limit delineated in MIL-STD-461.

3.7.3 Humidity. - During and after testing as specified in 4.13.3 the transmitter shall meet the performance requirements of 3.6.1.2 and 3.6.1.3.

3.7.4 Dust. - After testing as specified in 4.13.4 the transmitter shall meet the performance requirements of 3.6.1.2 and 3.6.1.3.

3.7.5 Altitude (elevation). - The transmitter shall meet the peformance requirements of 3.6.1.2 and 3.6.1.3 during testing as specified in 4.13.1.

3.7.6 Vibration. - The transmitter shall meet the performance requirements of 3.6.1.1, 3.6.1.2 and 3.6.1.3 both before and after testing as specified in 4.13.6.

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3.7.7 Leakage (immersion). -

3.7.7.1 Immersion, transmitter. - The transmitter shall show no evidence of leakage after it is tested as specified in 4.13.2.1.

3.7.7.2 Immersion, case. - The case shall show no evidence of leakage after it is tested as specified in 4.13.2.2.

3.7.8 Salt fog. - After testing as specified in 4.13.7 the transmitter shall meet the performance requirements of 3.6.1.2 and 3.6.1.3.

3.7.9 Fungus. - Upon completion of test, there shall be no deterioration, corrosion or change in configuration that will prevent the maintenance or operation of the transmitter. (See 4.13.8)

3.8 Burn-in. - Each transmitter T-1255/T shall be capable of meeting the performance requirements specified herein after being subjected to the specified burn-in cycle of 4.12.

3.9 Workmanship. - Each transmitter group shall be manufactured and assembled in accordance with the requirement 9 of MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may utilize his own facilities, facilities of a subcontractor or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that the supplies and services conform to prescribed requirements.

4.2 Classification of inspections. - Inspections shall be classified as follows:

a. First article inspection. (Does not include preparation for delivery) (See 4.3)

b. Inspections covered by subsidiary documents. (See 4.4)

c. Quality conformance inspections.

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(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.5)

(2) Quality conformance inspection of preparation for delivery. (See 4.7)

4.3 First article. - Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor.

4.3.1 First article units. - The contractor shall furnish the number of first article units in accordance with the following

a. Four (4) each Transmitter, Simulated Radiac, Hot Spot T-1255/T.

b. One (1) each Antenna Assembly AS-2919/T.

c. One (1) each Case, Simulated Transmitter CY-7349/T.

4.3.2 First article inspection. - The first article inspection shall consist of the inspections specified in Table II, and shall be performed in the order specified herein.

4.3.3 First article data. - The first article test plan and test report(s) shall be as required in the contract or purchase order.

Table II First article inspection. -

Inspection	Reqt Para	Test Para	Unit 1	Unit 2	Unit 3	Unit 4
1. Inspection covered by subsidiary documents	3.3	4.4	Inspections to be performed on all units			
2. Burn-in	3.8	4.12	Inspections to be performed on all units			
3. Group A inspection	See Table III		Inspections to be performed on all units			
4. Group B inspection	See Table IV		Inspections to be performed on all units			
5. EMI	3.7.2	4.14	1			

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Table II First article inspection. - (cont.)

Inspection	Reqt Para	Test Para	Unit 1	Unit 2	Unit 3	Unit 4
6. Enviromental						
Low temperature	3.7.1.1	4.13.5.1		1		
High temperature	3.7.1.2	4.13.5.2		2		
Humidity	3.7.3	4.13.3		4		
Dust	3.7.4	4.13.4				2
Altitude	3.7.5	4.13.1		3		
Vibration	3.7.6	4.13.6				1
Immersion	3.7.7	4.13.2	2			
Salt fog	3.7.8	4.13.7				3
Fungus	3.7.9	4.13.8			1	

NOTES:

1. The inspection 1 to 4, in the order shown, shall be performed on all first article units before subjecting these units to any other inspection requirements specified in the table.

2. The implementation of inspections 5 and 6 shall be expedited to satisfy contract delivery requirements.

3. The numbers in the unit columns in the table covered by 5 and 6 specify the order of inspections for the indicated unit.

4.4 Inspections covered by subsidiary documents. - The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification, and the inspection requirement specified in the contract or purchase order.

Item	Where required
Parts, Materials and processes-general	3.3
Marking	3.4
Workmanship	3.9

4.5 Quality conformance inspection of equipment before preparation for delivery. - The contractor shall perform the inspections specified in 4.4, 4.5.1, and 4.5.4 through 4.12. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

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4.5.1 Group A inspection. - Each unit on contract or purchase order shall be inspected for conformance to the inspections specified in Table III. Discrete lots shall be formed from units that passed this inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels, and AQLs indicated in Table III.

4.5.1.1 Order of inspection within Group A. - Group A inspection shall be performed in the order listed in Table III.

Table III Group A inspection.-

Inspection	Requirement Paragraph	Test Para	AQL Major	AQL Minor
Visual and Mechanical	3.9	4.15	2.5%	4.0%
Electrical				
Operating Frequency and Stability	3.6.1.1	4.11.1	2.5% for the	**
Power Output Stability	3.6.1.2	4.11.2	group	**

**All electrical defects are major.

4.5.2 Group B inspection. - Group B inspection shall be performed in inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and meet the Group A inspection. This inspection shall conform to Table I and to the special inspection levels of the Table I of MIL-STD-105.

4.5.2.1 Group B sampling plans. - The Group B sample plans, for the AQLs listed in Table IV, shall be as follows

AQL	Inspection level
4.0 percent	S-4
6.5 percent	S-4

4.5.2.2 Order of inspection within Group B - Group B inspection shall be performed in the order listed in Table IV.

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Table IV Group B inspection. -

Inspection	Requirement Paragraph	Test Para	AQL
Electrical			
Interchangeability	3.5	4.6	4.0%
Operating Frequency Range	3.6.1.1	4.11.1 1	4.0%
Operation in different modes	3.6.1.3	4.11.3	for
Weight (transmitter)	3.6.1.4	4.8.1	the
Battery life	3.6.1.5	4.9	group
Beam width (antenna)	3.6.2.1	4.10	
Weight (case)	3.6.3.1	4.8.2	
Immersion (case)	3.7.7.2	4.13.2.2	4.0%

4.5.3 Group C inspection. - Group C inspection shall be performed on units that have passed Group A and Group B inspection. The inspection shall consist of the inspections specified in Table V. Samples shall be selected in accordance with 4.5.3.1.

Table V Group C inspection. -

Inspection	Requirement Paragraph	Test Para
Subgroup 1		
Dust	3.7.4	4.13.4
Altitude	3.7.5	4.13.1
Subgroup 2		
Vibration	3.7.6	4.13.6
Fungus*	3.7.9	4.13.8
Subgroup 3		
Low temperature	3.7.1.1	4.13.5.1
High temperature	3.7.1.2	4.13.5.2
Leakage (immersion) (Transmitter)	3.7.7.1	4.13.2.1

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Table V Group C inspection.-(cont.)

Inspection	Requirement Paragraph	Test Para
Subgroup 4 Humidity	3.7.3	4.13.3
Subgroup 5 Salt fog*	3.7.8	4.13.7
Subgroup 6 EMI	3.7.2	4.14

*The Transmitter T-1255/T shall be thoroughly washed, cleaned, dried and refurbished after this inspection before proceeding with subsequent inspection.

4.5.3.1 Sampling for Group C inspection.-

4.5.3.1.1 Six samples of the complete Transmitter T-1255/T shall be selected at random, one sample for each Subgroup 1 through Subgroup 6 in Table V from every (production rate basis) or fraction thereof produced. The samples shall be selected at the start of the contract from the first quality conformance inspection lot. These samples shall constitute the Group C requirement for first twenty units produced. Subgroup 6 EMI, and Subgroup 2 fungus, inspections shall be performed only on first production lot.

4.5.3.2 Order of inspection within Group C. - Group C inspection shall be performed in the order specified in Table V.

4.5.3.3 Group C failures - Actions required relative to Group C failures shall be as specified in the contract or purchase order.

4.5.3.4 Reinspection of conforming Group C sample units. - Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass Group B and Group A inspection in that order.

4.6 Inspection for dimensional interchangeability. - The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3.5. When a listed dimension is not within specified design limits, it shall be considered a major defect.

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a. External and internal dimensions of cases, covers, and insertable assemblies, when such dimensions affect mating of parts.

b. Dimensions of cavities, when such dimensions affect insertion of items.

c. Location of hinges and fasteners on separable parts or assemblies which must mate, such as cases, covers, and mountings.

d. Location of connectors, locking pins, fasteners, slides, and mountings which receive mating parts of plug-in assemblies and major units; and location of the mating parts on the plug-in assembly or major unit.

e. Size and form of special threads.

4.7 Quality conformance inspection of preparation for delivery. - Preparation for delivery shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of Section 5.

4.8 Weight. -

4.8.1 Weight (Transmitter T-1255). - Using a calibrated scale the transmitter shall meet the requirement of 3.6.1.4.

4.8.2 Weight (case assembly). - Using a calibrated scale when fully loaded, the case shall meet the requirement of 3.6.3.1.

4.9 Battery life test. - Place the transmitter in the test chamber and stabilize the ambient temperatures to 0 ± 2.5°F. When the temperature of the transmitter stabilizes, operate for 12 hours in the CW mode from its internal batteries. (See 3.6.1.5)

4.10 Antenna beamwidth. - The beamwidth of the antenna shall be determined by a field strength meter in an actual field application. The beamwidth in degrees shall be determined as measured 3dB down from maximum reading obtained. The beamwidth shall meet the requirement of 3.6.2.1.

4.11 Performance characteristics. -

4.11.1 Operating frequency and stability. - Using the proper crystal for a frequency of 141 MHz, the Transmitter T-1255/T shall meet the requirement of 3.6.1.1. The operating frequency and stability requirements shall be met at maximum and minimum power output and at the minimum, nominal and maximum input voltages listed below. (See 3.6.1.1)

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	INPUT VDC ± 5%		
Minimum	14.4	7.2	4.5
Nominal	18.0	9.0	6.0
Maximum	19.8	10.8	7.2

4.11.1.1 Operating frequency range. - Using the methods and criteria of 4.11.1, the transmitter shall be tested at 138 MHz and 144 MHz. (See 3.6.1.1)

4.11.2 Power output and stability. - The power output and stability shall be tested in accordance with the following steps. (See 3.6.1.2) A dummy load impedance of 50 ohms shall be used.

Step 1. - With the transmitter set a maximum power and input voltages as specified in paragraph 4.11.1. All power output readings shall be between 225 to 275 mW.

Step 2. - Repeat Step 1 with the transmitter set at minimum power. All readings shall be less than 20 mW.

4.11.3 Operation in different modes. - In order to reduce the time of operation for testing (from 30 minutes to 5 seconds), a 2.0 K Ohm speed-up resistor (minimum 1/2 watt) should be placed between test points TP3 and TP4 on the timer subassembly board (refer to Drawing SM-E-748127).

4.11.3.1 Mode 1. - With the mode selector switch positioned to Mode 1, set the ON TIME and OFF DELAY switches as shown in Table VI and record the appropriate times for each setting. Time for each event shall be ± 20% for time listed in Table VI. Prior to each event the ON-OFF switch must be reset. (See 3.6.1.3.1)

Table VI Mode 1 Time-OFF DELAY/ON. -

OFF-DELAY	TIME ±20%	ON-TIME	TIME ±20%
.5hrs.	5 sec.	1 hr.	10 sec.
1.5hrs.	15 sec.	2 hrs.	20 sec.
2.5hrs.	25 sec.	3 hrs.	30 sec.
3.5hrs.	35 sec.	4 hrs.	40 sec.
4.5hrs.	45 sec.	5 hrs.	50 sec.
5.5hrs.	55 sec.	6 hrs.	60 sec.

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4.11.3.2 Mode 2. - Set the mode selector switch to Mode 2 position. Connect a wattmeter to the Transmitter T-1255/T output connector. Select from Table VI an OFF DELAY time. Turn the transmitter ON and within 6 seconds the wattmeter should indicate power output. The power shall increase to +24 dbm ± 1 db and then decrease to minimum power, at which time the transmitter shall turn off. The arrival/decay times shall be in accordance with 3.6.1.3.2.

4.11.3.3 Continuous Wave Mode. - With the mode selector switch in the CW position the transmitter shall meet the requirement of 3.6.1.3.3.

4.12 Burn-in. - Each transmitter submitted for acceptance shall be subjected to burn-in in accordance with Drawing SM-A-678029. (See 3.8)

4.13 Environmental. - The Transmitter T-1255/T shall be tested in accordance with the following environmental conditions.

4.13.1 Altitude (elevation) test. - The transmitter shall be tested in accordance with Method 500, Procedure 1, of MIL-STD-810. Prior to testing the transmitter shall meet the requirements specified in 3.6.1.1, 3.6.1.2, and 3.6.1.3. During tests of 4.11.2 and 4.11.3.

4.13.2 Leakage (immersion). -

4.13.2.1 Immersion, transmitter. - The transmitter shall be tested in accordance with Method 512, Procedure I of MIL-STD-810. (See 3.7.7.1)

4.13.2.2 Immersion, case. - The case shall be tested in accordance with Method 512, Procedure I of MIL-STD-810. (See 3.7.7.2)

4.13.3 Humidity (moisture-resistance) test. - The transmitter shall be tested in accordance with Method 507, Procedure II of MIL-STD-810. Prior to testing, the transmitter shall meet the requirements specified in 3.7.3. During the periods shown on Figure 507-2 of MIL-STD-810 perform the tests of 4.11.2 and 4.11.3. Following completion of test, perform the tests of 4.11.

4.13.4 Dust test. - The transmitter shall be tested in accordance with Method 510, Procedure I of MIL-STD-810. Prior to testing the transmitter shall meet the requirements specified in 3.7.4. Following completion of the test, perform the tests of 4.11.2 and 4.11.3

4.13.5 Temperature tests. -

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4.13.5.1 Low temperature procedure. - The following test shall be performed on the transmitter in accordance with MIL-STD-810, Method 502, Procedure 1. Perform the tests of 4.11.2 as pre-environmental data. Place the transmitter in the temperature chamber and reduce the ambient temperature to -60 ± 2.5°F. The test will start when the ambient temperature has stabilized. Hold the temperature for a period of 8 hours. At the end of 8 hours increase the ambient temperature to 0 ± 2.5°F. When the temperature of the transmitter has stabilized operate for six hours. During this time perform the tests of 4.11.2. Return the transmitter to room ambient temperature. Perform the tests of 4.11.2 as post environmental data. The readings at low temperature shall not vary more than ± 20% from those readings at room temperature and corresponding input voltage levels. (See 3.7.1.1)

4.13.5.2 High temperature procedure. - The following test shall be performed on the transmitter in accordance with MIL-STD-810, Method 501, Procedure 1. Perform the pre-environmental electrical performance test 4.11.2. Place the transmitter into the temperature chamber and increase the ambient temperature to 160 ± 2.5°F. The test will start when the ambient temperature has stabilized. Hold the temperature for a period of 8 hours minimum. At the end of 8 hours reduce the ambient temperature to 125 ± 2.5°F. When the temperature of the transmitter has stabilized operate for six hours. During this time perform test 4.11.2. Return the transmitter to room ambient temperature. Perform the tests of 4.11.2 as post environmental data. The readings at high temperature shall not vary more than ± 20% from those readings at room temperature and corresponding input voltage levels. (See 3.7.1.2)

4.13.6 Vibration. - The case, fully loaded, shall be vibrated in accordance with MIL-STD-810, Method 514.1, Procedure X, Curve AW and AY of Figure 514.1-7. Pre and post testing shall be performed on the transmitter in accordance with 4.11. (See 3.7.6)

4.13.7 Salt fog. - The transmitter shall be subjected to the salt fog test of MIL-STD-810, Method 509, Procedure 1. Pre and post data shall be performed in accordance with 4.11.2 and 4.11.3. (See 3.7.8)

4.13.8 Fungus. - The transmitter (including accessories) shall be tested in accordance with Method 508, Procedure 1 of MIL-STD-810. Spraying shall include parts and material on and under the chassis, and the interior of the equipment case. The transmitter shall remain in an open condition during the test. Operation is not required. (See 3.7.9)

4.14 EMI. - The transmitter shall be tested in accordance with MIL-STD-462 and meet the requirements of 3.7.2.

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4.15 Visual and mechanical inspection. - Equipment shall be examined for the defects listed in MIL-STD-252.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. - Preservation and packaging shall be level A or C, as specified. (See 6.2.b)

5.1.1 Level A. -

5.1.1.1 Cleaning. - Transmitter Group, Simulated Radiac, Hot Spot OT-52/T shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. - Transmitter Group, Simulated Radiac, Hot Spot, OT-52/T shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservation application. - None required.

5.1.1.4 Unit packaging. - Unit packaging shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical literature. - Each technical literature shall be packaged Method 1C-1.

5.1.1.4.2 Spare fuses. - Spare fuses shall be packaged Method III as follows: Cushion the fuses by wrapping together in paperboard conforming to PPP-P-291, Type 1. Secure the wrap with gummed paper tape conforming to PPP-T-45, Type III, Grade A.

5.1.1.4.3 Spare lamps. - Each spare lamp shall be individually packaged Method III as follows Cushion each lamp by wrapping in paperboard conforming to PPP-P-291, Type 1. Secure the wrap with gummed paper tape conforming to PPP-T-45, Type III, Grade A.

5.1.1.4.4 Transmitter Group, Simulated Radiac, Hot Spot. - Transmitter Group, Simulated Radiac, Hot Spot OT-52/T shall be individually packaged Method III as follows Stow all units within the designated compartments of transit case, including the items packaged as specified in 5.1.1.4.2 and 5.1.1.4.3. Close cover and secure all fasteners. Cushion the transit case on all four corners with corner pads fabricated of fiberboard conforming to PPP-F-320, Type CF, class weather-resistant, variety SW, Grade W5c, designed to protect all projections and absorb the shock of impact in

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handling and transit. Secure the corner pads to the transit case with tape conforming to PPP-T-76. Place the cushioned unit within a close-fitting fiberboard box conforming to PPP-B-636, W5c. Box closure shall be in accordance with the appendix of the box specification.

5.1.2 Level C. - Transmitter Group, Simulated Radiac, Hot Spot OT-52/T, shall be preserved and packaged in a manner that will afford adequate protection against physical and environmental damage during shipment, handling and limited intransit storage.

5.2 Packing. - Packing shall be level A, B or C as specified. Shipping containers for all levels shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents. (See 6.2.b)

5.2.1 Level A. - Transmitter Group, Simulated Radiac, Hot Spot OT-52/T, packaged as specified in 5.1, shall be individually packed within a close-fitting box conforming to PPP-B-601, overseas type, PPP-B-621, Style 4, Class 2, or PPP-B-585, Style 2 or 3, Class 3. Fiberboard boxes being placed directly in the shipping container shall be waterproofed with tape conforming to PPP-T-76, in accordance with the taping requirements of the appendix of the box specification. Place the technical literature, packaged as specified in 5.1.1.4.1 on top of the contents directly under the lid of the box. Skids, 3 X 4 inches, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that the metal strapping shall conform to QQ-S-781, Type 1, finish "A".

5.2.2 Level B. - Transmitter Group, Simulated Radiac, Hot Spot OT-52/T packaged as specified in5.1, shall be individually packed within a close-fitting box conforming to PPP-B-601, domestic type, PPP-B-621, Style 4, Class 1, or PPP-B-585, Style 2 or 3, Class 3. Skids, 3 X 4 inches, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specication, in a manner which will adequately support the item and facilitate the use of material handling equipment. Place the technical literature, packaged as specified in 5.1.1.4.1, on top of the contents, directly under the lid of the box. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto.

5.2.3 Level C. - Transmitter Group, Simulated Radiac, Hot Spot OT-52/T, packaged as specified in 5.1, shall be packed in shipping containers that comply with

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federal or military specifications, in a manner that will afford adequate protection to the package and its contents against physical and environmental deterioration and damage, during limited shipment, handling and intransit storage.

5.3 Marking. - In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. - The Transmitter Group, Simulated Radiac, Hot Spot OT-52( )/T used with the Receiver Group, Simulated Radiac OR-114/T comprises a trainer system which allows for safe instruction and practical exercise in the survey and mapping of radiological contamination.

6.2 Ordering data. - Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.

b. Levels of preservation and packaging and packing. (See Section 5)

c. When rough handling and functional tests are required.

d. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

e. Marking and shipping of samples.

f. Place of final inspection.

g. Transmitter is powered by five BA-409/U batteries that meet requirements of MIL-B -18. The Government will not procure batteries with Transmitter Group OT-52( )/T.

Custodian:
Army-EL

Preparing Activity:
Army - EL
Project No. 6665-A316

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 22-R255

INSTRUCTIONS This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated Fold on lines on reverse side staple in corner and send to preparing activity Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements

SPECIFICATION MIL-T-55402(EL) TRANSMITTER GROUP, SIMULATED RADIAC, HOT SPOT OT-52( )/T

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO (If "yes" in what way?

4 REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity Optional

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

ESC-FM 1068-68

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